Daily Drilling Report
SPD19B-03
9/5/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
40.0	8 1/2"	185.00	72.83	31.70	92.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	220,877.90	11,064.63	9,532,082.82		35.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,725.00	2,756.00	2,609.25	31.00	10.10	3.1

Operations at Report Time
Drilling @ 2780m.

Operations Summary
RIH directional BHA#10 and tag plugs @ 2696m. D/O shoe track and rathole to 2725m. Drill new formation to 2728m. Condition the mud. Perform FIT. Drilling 8 1/2" hole section f/ 2728m to 2756m.

Operations next 24hrs
Cont' drilling 8 1/2" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari / Behnam Jahangiri	Night Drilling Sup.
Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/31/2014	10/1/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/5/2014	10/21/2014
Trip Drill	9/4/2014	10/25/2014

STOP Cards

Number of Reports: 41

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	03:30	3.50	8 1/2"	D	04	P	RIH dir. drilling BHA#10 on 5 7/8" DP f/ 1900m to 2664m. Connect TDS, W/D w/ 300gpm / 1000psi, tag top cement plug @ 2696m.
03:30	06:00	2.50	8 1/2"	D	03	P	D/O CMT plugs + Float collar (2696.5m) + cement inside shoe track + Float shoe (2720.8m) and clear rathole to 2725m w/ 450 gpm / 1850 psi / 2 - 10 WOB / 30 rpm / 14 - 18 Kft-lbf. *** Pump 30 bbls Hi-Vis pill (from old barite mud) after D/O float collar. *** Check shoe with and without rotation, OK. *** Pump 50 bbl Hi-Vis pill + 100 bbl SW + 650 bbl new CaCo3/KCL/ Polymer 10.5 ppg mud and displace hole while D/O cement.
06:00	07:00	1.00	8 1/2"	D	03	P	Drilling 3m of new formation 8 1/2" hole f/ 2725m to 2728m w/ 500 gpm / 1850 psi / 15 kips / 30 rpm / 15-20 TRQ.
07:00	08:00	1.00	8 1/2"	D	07	P	Condition the mud by circulating w/ 450 gpm / 1450 psi/ 15 rpm. Meanwhile, keep the string in motion. (MW in and out: 10.5 ppg)
08:00	08:30	0.50	8 1/2"	D	11	P	Pull string inside 9 5/8" shoe. Line up cement unit to choke manifold. Pressure test lines to 500 / 2000 psi , 5 / 10 min , OK. Close upper pipe ram. Perform FIT to 14 ppg EMW w/ 10.5 ppg CaCo3/KCL/Polymer mud by applying 1540 psi surface pressure for 10 min, Ok. Pump totally 5.7 bbls and get same volume in return.
08:30	00:00	15....	8 1/2"	D	03	P	Directional drilling 8 1/2" hole section in rotation and slide mode f/ 2728m to 2756m w/ 500-550 gpm / 1800-2300 psi / 15-25 kips/ 30-50 rpm / 15-20 Kft-lbf. Rotation Interval Slide Interval 2725m - 2753m 2753m - 2756m ** Drilling with NL/NG. ** B/R & R/D each stand drilled once. ** Take survey before connection.

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Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00						Sep 06, 2014 00:00-06:00 Cont' Directional drilling 8 1/2" hole section in rotation and slide mode f/ 2756m to 2780m w/ 550 gpm / 2200-2300 psi / 25 kips / 50 rpm / 15-20 Kft-lbf. Slide Interval Rotation Interval 2756m - 2765m 2765m - 2780m ** Last sample @ 2775m: (40% Dol.) + (30% Lst.) + (30% Chky Lst.)

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,756.00	CaCo3/KCL/ Polymer	10.50	54	28.0	22.000	3.0
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.5	0.300	1.100	1.200			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
4	6	0.0	86.0	14.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
5.000	6.000	8.000	0/100	53.0	1449.0	3080.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
6495.0	25.0	0.0	0.01		9.050506	0.499

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.50	6.	300	10.0	996.0	37.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Swissco	Supply Boat	9/5/2014 15:40	9/6/2014 03:35	Delivered 15 BB of CACO3 F & Brine-730 BBLs f/ Aban-8. Received 700 BBLs of Mud, departed for SR1.
C Mariner	Supply Boat	9/5/2014 15:45	9/6/2014 00:55	Arrived f/SHJ. Delivered Cargo, FO & FW. Received Cargo for ABAN8. Departed to ABAN8.

Mud Additives

Description	Consumed	Cum Rec	On Loc
CaCo3-F BB	4.0	174.0	68.0
Caustic Soda 2014	5.0	500.0	311.0
Duo-Tec	10.0	185.0	121.0
Starch	80.0	1,600.0	417.0
Pac-LV	27.0	521.0	78.0
PTS (Temp. Stabilizer)	2.0	19.0	12.0
Soda Ash 25 Kg	10.0	750.0	367.0
KCL	2.0	52.0	19.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	23.0	152.0	203.0
Drill Water	10.0	10.0	330.0
Diesel Fuel	11.0	50.0	199.0
Cement G	0.0	0.0	92.0
Barite Bulk	0.0	0.0	129.0
Barite BB	0.0	0.0	0.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
10	8 1/2	Varel	VM713GUX	6006867	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	10.10	117,111.00	2,725.00	2,756.00	31.00	3.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
500	550	1,825.0	2,330.0	15	25	160	210
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
18.90	9.27	91.78	8.22	416	352	285	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,725.00	2,736.00	2.27	18,000.0	500	30	1,825.0	15	341
2,736.00	2,739.50	2.41	18,000.0	550	30	2,150.0	15	341
2,739.50	2,753.00	4.59	18,000.0	550	50	2,180.0	25	352
2,753.00	2,756.00	0.83		550		2,330.0	25	

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.89	255.95
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	228.06
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	222.26

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Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
X/O	6 1/2	2.813	0.85	35.91
6 3/4" Float sub	6 3/4		0.81	35.06
6 3/4" NMDC	6 3/4	3.250	9.48	34.25
6 3/4" TM - Sub	6 3/4		1.74	24.77
6 3/4" NMDC	6 3/4	3.250	9.48	23.03
8" St. Stabilizer (COR-2491AD)	8	3.000	1.84	13.55
Pony NMDC	6 13/16	3.500	3.09	11.71
6 3/4" Motor (AKO: 1.15° , Sleeve: 8 3/8")	6 3/4		8.34	8.62
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
2	2,753.00	14.00	20	240.0	97	114
2	2,753.00	14.00	30	360.0	97	170
2	2,753.00	14.00	40	540.0	97	227
3	2,753.00	14.00	40	550.0	97	227
3	2,753.00	14.00	20	230.0	97	114
3	2,753.00	14.00	30	360.0	97	170

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,729.12	249.40	19.40	2,584.06	-341.55	-748.58	822.82	1.85	-1.84
2,758.26	249.10	21.60	2,611.35	-345.17	-758.12	833.00	2.27	2.26

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S3		2,622.00	2,483.90
S2		2,632.50	2,493.72
S1		2,675.70	2,534.06
Upper Kangan		2,703.00	2,559.14
K1		2,732.00	2,586.77

General Remarks

Comment

>> Requirements: (200 sxs Starch f/ MR #14 + MI MR #15)

>>> Sent:

** C Mariner to ABAN-8: (8 jts 9 1/2" Drill Collar)

** Swisco Supplier to SR1: (700 bbls KCL/Polymer/Glydrill/Barite mud)

>>> Received:

** C Mariner f/ Sharjah: (ABAN Equipment) + (ABAN 48 jts 5 7/8" DP & 4 jts 5 7/8" HWDP) + (400 sxs Starch + 40 sxs Pac UL + 200 sxs Soda Ash + 160 sxs Caustic Soda) + (50 MT FO) + (150 MT FW)

** Swisco Supplier f/ABAN-8: (15 BB CaCo3-F) + (730 bbls brine)